Extraction of Microservices from Monolithic Software Architectures

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Paper Information: Key Paper

Rating: ⭐ ⭐ ⭐ ⭐

Own Keywords: Approaches, Clustering, Graph-based

Reading Progress: Second Pass

Introduction (What is the Paper about?)

- addressing the decomposition problem of monolithical applications into microservices **algorithmically and graph-based**
- apply approaches from the software decomposition research formally and (semi-)automated to microservices

Methods

- constructed a web-based prototype
- tested three formal coupling strategies used to construct a graph from code
  - perform a clustering algorithm on that
- performance evaluation on the prototype with custom microservice-specific metrics on 21 open source projects in Java, Ruby and Python

Scientific Benefit

- Presenting a formal microservice extraction model as web-based prototype
  - semi-automated without heavy user input
- allow algorithmic recommendations of microservice candidates in a refactoring and migration scenario
- size of produced microservice candidates conforms with reported microservice size in empirical surveys
- produced microservice candidates lower the average development team size down to half of the original size or lower
- domain-specific redundancy among different microservices is kept low

Challenges / Future Work

Limitations

- extraction model uses classes as atomic unit rather than methods or procedures
  - left for future work
- extraction model treats database entities as normal classes
  - still unsolved to share or assign pre-existing databases to different models

Paper Outline and Read Progress

1. [X] Introduction
2. [X] Related Work
3. [ ] Formal Definition of the Extraction Models, Coupling Strategies and Clustering Algorithm
4. [ ] Performance and Quality Evaluation
5. [X] Limitations and Future Work
6. [ ] Conclusion

Own Comments

- (+) Potential key paper
- (+) First Paper applying software decomposition methods to microservices
- (-) potentially outdated
- (-) custom metrics
- (-) missing reference of Komondoor et al.